

Wei-Hsuan Chung 鍾瑋軒

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RESEARCH PROFILE

Senior undergraduate in physics at National Taiwan University whose research spans waveguide quantum electrodynamics, open quantum systems, and quantum error correction. Experienced in theoretical modeling and numerical simulation using Python and Julia. Co-author of a peer-reviewed article published in *Physical Review A*, with research interests at the intersection of theoretical quantum information and quantum optics.

EDUCATION

- **National Taiwan University (NTU)** Sep. 2023 - Jun. 2027 (Expected)
B.S. candidate in Physics; double major in Sociology Taipei, Taiwan
 - CGPA: 3.99/4.30
 - Completing the **NTU Quantum Computing and Information Program**

RESEARCH EXPERIENCE

- **Max Planck Institute for the Science of Light (MPL)**  Sep.–Oct., 2026
Research Intern, Lise Meitner Research Group led by Dr. Flore Kunst Erlangen, Germany
 - Selected for an eight-week research internship in the Non-Hermitian Topological Phenomena group.
 - Will conduct theoretical research on non-Hermitian and topological phenomena in open quantum systems using analytical and numerical methods.
- **Max Planck Institute of Quantum Optics (MPQ)**  Aug. 2026
Summer Research Intern, Theoretical Group, supervised by Dr. Rahul Trivedi Garching, Germany
 - Selected as one of 13 participants from a pool of 386 applicants for the 2026 MCQST Summer Bachelor Program.
 - Conducting a research project entitled *Robustness of Error-Correction Codes to Realistic Noise*.
 - Investigating the performance of quantum error-correcting codes under realistic noise models with simulations.
 - Using Python and Stim to simulate quantum errors, syndrome measurements, and error-correction procedures.
- **Institute of Atomic and Molecular Sciences (IAMS), Academia Sinica**  Jul. 2025– Present
Research Assistant, Atomic Physics and Optical Science Theory Group, Dr. Hsiang-Hua Jen Taipei, Taiwan
 - Joined the group through the 2025 IAMS Undergraduate Summer Internship and continued as a research assistant.
 - Performed numerical simulations of excitation dynamics and spectral properties in structured waveguide QED using Python and Julia; contributed to simulation, analysis, and manuscript writing for a *Physical Review A* publication.
 - Investigating Dicke-state generation through quantum-coherence control in directional waveguide QED; manuscript in preparation.
 - Studying non-equilibrium quantum phenomena, including the Quantum Mpemba Effect and quantum batteries, in waveguide and cavity QED systems.
- **Institute of Physics (IOP), Academia Sinica**  Aug. 2024–Jul. 2025
Research Assistant, Complex Systems of Granular Materials Group, Dr. Kiwing To Taipei, Taiwan
 - Conducted experiments on reciprocal-force behavior in granular systems and developed Linux shell scripts for experimental data processing and analysis.
 - Continued the project independently by developing LAMMPS simulations.

- Developed *Luminer*, an AI teaching assistant for introductory physics courses, using Python, Streamlit, and the Gemini API.
- Analyzed student–chatbot interactions to evaluate system performance and identify opportunities for improving educational effectiveness.
- Reviewed Just-in-Time Teaching (JiTT) literature and analyzed student feedback and performance data to assess its effectiveness in undergraduate physics education.

PATENTS AND PUBLICATIONS

C=CONFERENCE, J=JOURNAL, P=PREPRINT, S=IN SUBMISSION, T=THESIS

- [J.1] I. G. N. Y. Handayana, Y.-T. Yu, **W. H. Chung** and H. H. Jen, **Manipulating Excitation Dynamics in Structured Waveguide Quantum Electrodynamics**, *Physical Review A* **113**, 043709 (2026).; doi:10.1103/nx83-npj8.
- [P.1] **W. H. Chung***, Y.-T. Yu*, T.-S. Guo and H. H. Jen, **Fast Quantum State Preparation in Chiral Waveguide QED**, manuscript in preparation. (* Co-first author)
- [P.2] M.-Y. Liao*, **W. H. Chung*** and Pei-Yun Yang, **Teaching General Physics with Just-In-Time Teaching**, manuscript in preparation. (* Co-first author)

ACADEMIC ACTIVITIES

- **2026 IBM Quantum Dream: New York Training Camp** New York, USA
Taiwan Global Pathfinders Initiative, Ministry of Education Jul. 13–28, 2026
 - Selected for a fully funded overseas training program in quantum computing under the Taiwan Global Pathfinders Initiative.
 - Participated in lectures, hands-on workshops, and institutional visits covering quantum hardware, quantum algorithms, error correction, and applications in finance and emerging technologies.
 - Engaged with researchers and industry professionals from IBM Quantum, PsiQuantum, JPMorgan Chase, and Wells Fargo on the development and commercial applications of quantum technologies.
- **2026 Annual Meeting of the Physical Society of Taiwan** Chiayi, Taiwan
National Chung Cheng University Jan. 13–15, 2026
 - Presented a poster entitled *Manipulating Excitation Dynamics in Structured Waveguide Quantum Electrodynamics*.
- **2025 Workshop on Advances in Quantum Error Correction** Hsinchu, Taiwan
National Yang Ming Chiao Tung University Dec. 29, 2025
 - Attended lectures on QLDPC codes, surface codes, and recent advances in fault-tolerant quantum computing.
- **2025 NASA Space Apps Challenge** Taiwan
NASA Oct. 2025
 - Collaborated with a multidisciplinary team to develop a data-driven web application addressing global challenges.
 - Developed [WeAdvisor: Discover Global Historical Weather Data](#) using React . js and Flask.
- **2025 NCTS AMO Physics Summer School** Hsinchu, Taiwan
National Center for Theoretical Sciences, Physics Division Aug. 2025
 - Completed an intensive three-day program covering contemporary topics in atomic, molecular, and optical physics.

SELECTED PROJECTS

- **Noise-Aware Evaluation of EPOC Compilation** 2025
Introduction to Quantum Computation and Information
 - [Project Report](#)
 - Main Reference: [EPOC: An Efficient Pulse Generation Framework with Advanced Synthesis for Quantum Circuits](#)
 - Reproduced the EPOC compilation pipeline by combining ZX-calculus optimization with BQSKit synthesis and evaluated its performance under thermal-relaxation and depolarizing noise using Qiskit Aer.
 - Observed depth reductions exceeding 90% on selected benchmarks and identified degradation on larger circuits when pulse durations approached hardware coherence times.
- **Quantum Algorithms for Partial Differential Equations** 2025
Quantum Information and Computation
 - Project: [QIC Final Project \(Presentation Video, Project Report, and Slides\)](#)
 - Surveyed Hamiltonian-simulation, block-encoding, and finite-difference approaches to solving PDEs on quantum computers, comparing their resource scaling and implementation requirements.

- Analyzed practical limitations arising from state preparation, boundary conditions, observable extraction, and Trotter error.
- **Theoretical Foundations of Neural Scaling Laws** 2025
Physical Theories of (Machine) Learning
 - Main Reference: [Bahri, Y., Dyer, E., Kaplan, J., Lee, J., & Sharma, U. \(2021\). Explaining neural scaling laws. arXiv.](#)
 - Project: [My Presentation Slides](#)
 - Presented a literature review based on Bahri et al. on theoretical explanations of neural scaling laws across dataset- and model-size regimes.
 - Compared variance-limited scaling, arising from statistical fluctuations, with resolution-limited scaling governed by the intrinsic dimension and geometric structure of the data.

SKILLS

- **Programming:** Python, Julia, Linux shell scripting
- **Scientific Computing:** QuTiP, QuantumOptics.jl, NumPy, SciPy, Stim, Qiskit, LAMMPS
- **Research Tools:** Git, Jupyter Notebook, LaTeX, LabVIEW
- **Research Methods:** Numerical simulation, quantum-system modeling, stabilizer-circuit simulation, experimental data acquisition and analysis
- **Languages:** Mandarin (Native), English (CEFR C1), German (Beginner)

HONORS AND AWARDS

- **Honorable Mention, Undergraduate Research Poster Competition** May 2026
Department of Physics, National Taiwan University 
 - Awarded for the poster *Manipulating Excitation Dynamics in Structured Waveguide Quantum Electrodynamics*.
- **Merit Award, Promenade of Data Science 2024** Dec. 2024
Institute of Statistical Science, Academia Sinica 
 - Recognized for excellence in data science skills, innovative problem-solving and great teamwork
 - Using Python and machine learning for data analysis and visualization
 - Awarded among 100+ participants from various universities, including undergraduate and graduate students
- **Entegris Foundation STEM Scholarship Fund** 2024
Entegris Foundation, Inc 
 - Selected for academic excellence and commitment to STEM from among sophomore applicants at National Taiwan University
- **Mr. Cheng-Chih Yen Memorial Scholarship** 2025
Department of Physics, National Taiwan University 
 - Awarded by the Department of Physics during the Fall 2025 semester